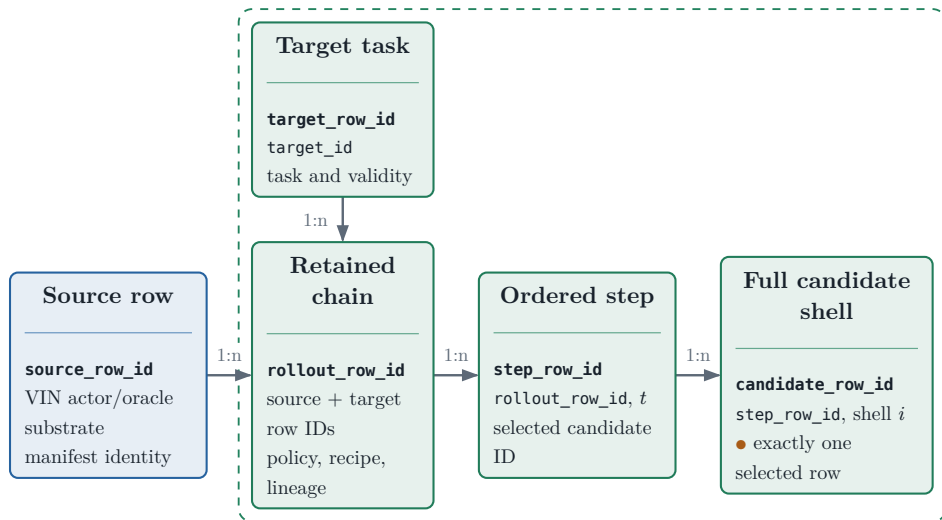
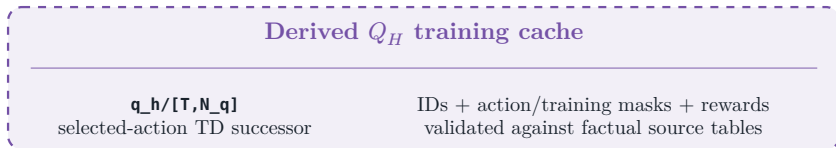




materialize: join IDs → copy/validate masks → pad



Scientific boundary. Solid links preserve factual row references; ● marks the selected candidate. The dashed Q_H cache is a verified projection of those rows, not an independent transition table.